ZEPHIRUM · ZYQL

A LANGUAGE THAT REFUSES TO GUESS. EVERY TIME.

ZEPHIRUM is the language. ZYQL is the algorithm.

Every decision comes out signed: 1 · 0 · REFUSED.

HELLO WORLD — EXECUTED FOR REAL

```
print "Hello, World!"
```

```
Hello, World!
```

exit 0 · real interpreter · nothing faked

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ZEPHIRUM

A native language with quantum primitives




```
print "Hello, World!"
```

the first program of the language

```
bell = qpu_run("bell", 400)
```

entangled pair, straight from the syntax

```
print "outcomes:", bell
```

measurement in one line

```
chi2 = 3.24 < 6.635 -> VERIFIED
```

statistical verdicts, not vibes

Bignums · functions · recursion · bit builtins · QPU bridge.

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ZYQL

The exact-decision gate

1	A question enters the gate.
0	Degenerate input is rejected.
REFUSED	Undecidable claims are refused — no guessing, ev

EVERY OUTPUT CARRIES A RECEIPT

Signed with ML-DSA-44. Tamper-evident.

A claim without a receipt is not a claim.

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The receipt battery

KAPLANSKY 90,112 pairs over F2 — 0 violations	MEASURED
AP-FREE SETS r3(N) N=8..16 — no 3-term progression	MEASURED
DIVISOR CORRELATION ratio extrapolates to 6/PI^2	MEASURED
BELL PAIR 96.2% Z / 97.5% X correlation	HARDWARE
GHZ STATE 96.3% full correlation	HARDWARE
CHSH S = 2.833 — local limit 2 violated	HARDWARE

Hardware runs via the QUANTUM HARDWARE BRIDGE (156-qubit class).

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What it does NOT claim

- No quantum advantage claimed. Ever.
- No vendor endorsement implied.
- No oracle prediction for markets or release dates.
- No measurement fabricated. SIMULATION MODE is labeled.
- Undecidable questions return REFUSED, not an opinion.
- Every receipt is reproducible or it does not ship.

A field matures when it stops measuring the machine and starts measuring the work.

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Deterministic, signed, portable

SHA-256
100% parity vs reference (FIPS 180-4)

SHA3-256
100% parity vs reference (FIPS 202)

GATE SELFTEST
3/3 green — signature valid, tampering caught

LEDGER AUDIT
38/38 receipts re-verified by the C core

SEMANTICS
naked question -> REFUSED · evidence -> 1

One C99 core. Same receipts on every machine.

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The fusion battery

Every external claim enters the same gate.

BOHR QUANTIZATION
H-alpha 656.47 nm from first principles — CONFIRMED

ENERGY CONSERVATION
H ground state 13.5983 eV, dev 0.00 — CONFIRMED

QUTRIT CASCADE
green 529.6 nm then red 650.2 nm — CONFIRMED

CHSH LOCAL LIMIT
S=2.833 vs classical 2 — VIOLATION MEASURED

FTL CLAIMS
refuted by measurement, no signaling found

FIREWORK CHEMISTRY
REFUSED — outside the model class

32 signed receipts across books, posts and live claims.

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Confirmed components only



The language, the kernel, the gate, the ledger. No vendor names.

Quantum primitives, plain text

hello.zeph

```
print "Hello, World!"
```

the first program

bell.zeph

```
bell = qpu_run("bell", 400)
```

entangled pair

ghz.zeph

```
ghz = qpu_run("ghz", 600)
```

genuine tripartite

kaplansky.zeph

```
pairs = kaplansky_f2(S3, Z6, Z7, Q8)
```

90,112 pairs, 0 violations

One syntax. Classical bignums and QPU circuits in the same file.

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ZYQL



1 · 0 · REFUSED — the exact-decision gate.

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The cyan betta



Exact-decision language: 1 / 0 / REFUSED.

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Superposition in code



The lemniscate: one loop, two states, no guessing.

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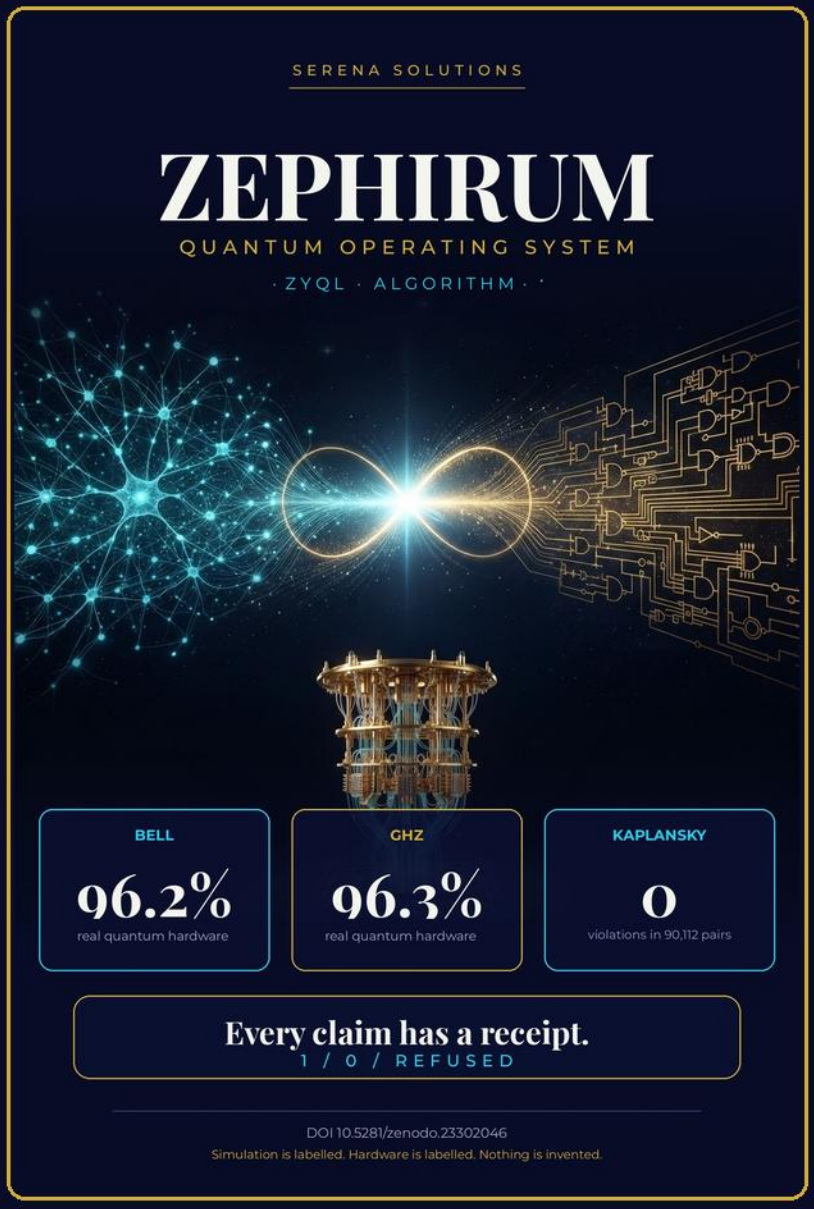
Neural net meets logic gate



Learning on the left. Exact logic on the right. A quantum core in between.

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Every claim has a receipt



Bell 96.2% · GHZ 96.3% · Kaplansky 0 violations in 90,112 pairs.

Chi-squared, verified

Every Claim Has a Receipt.

Quantum language that refuses to guess.

ZEPHIRUM is the language. ZYQL is the algorithm.



```
chi2 = 3.24 < 6.635  VERIFIED
```

ML-DSA-44 / FIPS 204 - DOI 10.5281/zenodo.23302046

SERENA SOLUTIONS

chi2 = 3.24 < 6.635 — ML-DSA-44 signed.

Prove before you pay



Q-SIM Gateway: evidence first, then the decision.

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EVERY CLAIM HAS A RECEIPT. YOURS TOO.

ZEPHIRUM is the language.
ZYQL is the algorithm.

RUN THE HELLO WORLD YOURSELF

```
zephirum.py run examples/hello.zeph
```

Question everything. Refuse what cannot be proven.

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